SYNTHETIC DEMO - All inputs and outcomes are illustrative.

Experiment calculation memorandum

Partner growth experiment

Build test plan | Owner: Growth team | Protocol v1

Experiment ID: customer-business-v2-demo

Protocol saved: 2026-10-26T21:08:06+00:00 | Snapshot calculated: 2026-10-26T21:08:06+00:00

Traceable calculation memorandum. This document does not establish completed audit or approval.

Decision and experiment definition

Does a simpler offer increase revenue without increasing cancellations?

Name	Role	Definition
Control	Control	Current offer
Test A	Test	Simplified offer
Test B	Test	Simplified offer with reminder

Unit	Decision Scope	Start Date	Window Days	Maturity Delay Days
Customer	Overall	2026-10-26	28	0

Unit	Eligible Per Week	Max Recruitment Weeks
Customer	2500	12

Metric definitions and decision thresholds

revenue - Primary / Continuous / Higher is better

Name	Unit	Definition	Window Days	Effect
revenue	currency / customer	Total per randomized customer over the observation window	28	5

Name	Effect Scale	Margin	Alert Threshold	Include In Plan
revenue	Relative %	0	0	True

cancellation - Guardrail / Binary / Lower is better

Name	Unit	Definition	Window Days	Effect
cancellation	rate (0 to 1)	Cancellation within 28 days per randomized customer	28	1

Name	Effect Scale	Margin	Alert Threshold	Include In Plan
cancellation	Percentage points	0.03	0.02	True

Population and frozen weights

Fixed-weight Partner x Segment mean difference; variance = sum(w^2 s^2/n); Welch-Satterthwaite df. Binary: fixed-weight risk difference with normal inference when adequate, exact per-cell fallback.

Partner	Segment	Weight
Partner A	New	0.25
Partner A	Existing	0.25
Partner B	New	0.25
Partner B	Existing	0.25

Design assumptions

Alpha	Power	Multiplicity	Attrition	Allocation
0.05	0.8	True	0	Recommended

Alpha	Assumption Only	Historical Start	Historical End
0.05	False		

Predeclared comparison family: 4. Planning alpha per comparison: 0.0125. Power is per planned comparison.

Arm	Partner	Segment	Metric	Source
Test A	Partner A	Existing	cancellation	Control variance x 1^2 (assumed Test SD multiplier)
Test A	Partner A	Existing	revenue	Control variance x 1^2 (assumed Test SD multiplier)
Test A	Partner A	New	cancellation	Control variance x 1^2 (assumed Test SD multiplier)
Test A	Partner A	New	revenue	Control variance x 1^2 (assumed Test SD multiplier)
Test A	Partner B	Existing	cancellation	Control variance x 1^2 (assumed Test SD multiplier)
Test A	Partner B	Existing	revenue	Control variance x 1^2 (assumed Test SD multiplier)
Test A	Partner B	New	cancellation	Control variance x 1^2 (assumed Test SD multiplier)

Arm	Partner	Segment	Metric	Source
Test A	Partner B	New	revenue	Control variance x 1^2 (assumed Test SD multiplier)
Test B	Partner A	Existing	cancellation	Control variance x 1^2 (assumed Test SD multiplier)
Test B	Partner A	Existing	revenue	Control variance x 1^2 (assumed Test SD multiplier)
Test B	Partner A	New	cancellation	Control variance x 1^2 (assumed Test SD multiplier)
Test B	Partner A	New	revenue	Control variance x 1^2 (assumed Test SD multiplier)
Test B	Partner B	Existing	cancellation	Control variance x 1^2 (assumed Test SD multiplier)
Test B	Partner B	Existing	revenue	Control variance x 1^2 (assumed Test SD multiplier)
Test B	Partner B	New	cancellation	Control variance x 1^2 (assumed Test SD multiplier)
Test B	Partner B	New	revenue	Control variance x 1^2 (assumed Test SD multiplier)

Sample requirements and timing

Recruitment targets include the saved missing-outcome reserve.

Arm	Analyzable N	Recruitment N
Control	2231	2231
Test A	2230	2230
Test B	2230	2230

Recruitment duration: 2.6764 weeks. Recruitment end: 2026-11-14. Earliest mature readout: 2026-12-12. Within maximum recruitment duration: True.

Partner	Segment	Arm	Analyzable N	Recruitment N
Partner A	New	Control	558	558
Partner A	Existing	Control	558	558
Partner B	New	Control	558	558
Partner B	Existing	Control	557	557
Partner A	New	Test A	558	558
Partner A	Existing	Test A	558	558
Partner B	New	Test A	557	557
Partner B	Existing	Test A	557	557
Partner A	New	Test B	558	558
Partner A	Existing	Test B	558	558
Partner B	New	Test B	557	557
Partner B	Existing	Test B	557	557

Partner	Expected Share
Partner A	0.33333333
Partner A	0.33333333
Partner B	0.33373206
Partner B	0.33333333
Partner A	0.33333333
Partner A	0.33333333
Partner B	0.33313397
Partner B	0.33333333
Partner A	0.33333333
Partner A	0.33333333
Partner B	0.33313397
Partner B	0.33333333

Pairwise Neyman and shared-control calculation

First solve each Test versus Control using the Primary metric SD ratio and all included metric constraints. Take the largest pairwise Control N. Fix it, solve each Test again, then verify all final integer stratum counts. Business-fixed allocation instead scales its declared ratios.

Scope	Test	Pair Control N	Pair Test N	Neyman Ratio
Overall	Test A	2231	2231	1
Overall	Test B	2231	2231	1

Scope	Metric	Test	Control N	Test N
Overall	revenue	Test A	2231	2230
Overall	revenue	Test B	2231	2230
Overall	cancellation	Test A	2231	2230
Overall	cancellation	Test B	2231	2230

Scope	Detectable Change Or Margin	Planning Alpha	Power	Variance Of Difference	Degrees Of Freedom
Overall	6.125	0.0125	0.99966712	1.0765749	4304.7006
Overall	6.125	0.0125	0.99966712	1.0765749	4304.7006
Overall	0.03	0.0125	0.80005821	8.0699456e-05	4452.9865
Overall	0.03	0.0125	0.80005821	8.0699456e-05	4452.9865

Assumptions, gaps and limitations

- Power uses plug-in historical moments and an approximate fixed-weight stratified Welch / normal model; historical parameter uncertainty is not fully propagated.

- Target power applies to each planned comparison, not to simultaneous success of every metric or arm.

- Pairwise Neyman with shared control is intuitive and verified, but is not a global minimum-total-sample optimization.

- Recruitment dates assume stable eligible traffic and adequate recruitment in every selected stratum.

- Fixed-horizon inference: repeated monitoring does not authorize early efficacy stopping.

- Binary planning assumes a common absolute risk difference across strata at the design alternative, or zero true difference for Guardrail non-inferiority power; rates are never silently clipped.

- The historical period was not fully supplied. Confirm the source is representative and uses the configured metric observation windows.

Technical appendix: formulas and complete inputs

Tables display eight significant digits. Companion JSON retains full numeric precision for independent recalculation.

- Independent randomization units; mutually exclusive Partner x Segment strata; fixed weights identical in every arm.

- Estimate = sum_h w_h (mean_Test,h - mean_Control,h). Variance = sum_h w_h^2 (s_Test,h^2/n_Test,h + s_Control,h^2/n_Control,h).

- Welch-Satterthwaite df = variance^2 / sum_h,arm [(w_h^2 s_h,arm^2/n_h,arm)^2/(n_h,arm-1)].

- Continuous planning: noncentral t approximation with the same integer cell counts and df. Binary planning and large-cell testing: fixed-weight risk difference normal approximation; sparse cells require exact procedures.

- Continuous Guardrail: benefit-oriented difference compared with -margin; non-inferiority p-value is one-sided. Decisions use conservative family intervals, not p > 0.05.

- Adequate Binary cells use fixed-weight normal risk-difference intervals with family protection. Sparse cells use Newcombe-Wilson cell intervals with additional Bonferroni protection over strata; Fisher p-values do not invert these intervals.

- SMD uses pre-treatment covariates only. SRM uses assigned counts and frozen stratum allocation. Missing checks are never passes.

- Historic sample variance uses ddof=1 / VAR.S. Missing outcomes are not zeros; observed outcomes are not assignment counts.

Partner	Segment	Metric	Arm	Sample Mean
Partner A	New	revenue	Control	100
Partner A	New	cancellation	Control	0.1
Partner A	Existing	revenue	Control	120
Partner A	Existing	cancellation	Control	0.1
Partner B	New	revenue	Control	125
Partner B	New	cancellation	Control	0.1
Partner B	Existing	revenue	Control	145
Partner B	Existing	cancellation	Control	0.1

Partner	Sample Variance	Sample Number	Successes
Partner A	900	1200	Not available
Partner A	0.090075063	1200	120
Partner A	1156	1200	Not available
Partner A	0.090075063	1200	120
Partner B	1225	1200	Not available
Partner B	0.090075063	1200	120
Partner B	1521	1200	Not available
Partner B	0.090075063	1200	120

Reproduction fingerprints

Python	Numpy	Pandas	Scipy	Statsmodels
3.11.15	1.26.4	2.2.2	1.13.1	0.14.2

Python	Streamlit	Reportlab
3.11.15	1.39.0	4.4.9

Code: customer-v2-sha256:22a48312d3977f1fa7ce93127e4d1d4909fe104bb255b0eb94104a9d03f2cf0a
Configuration: b224399aed519178fb3e87d8c66c355911e429557767faed2058115bba182b3d
Historical inputs: 6db1f0a135c8fb72cc56d20247cd1193da534f518d18d0d12c13555e438dbaec
Snapshot inputs: Not provided

Download the accompanying protocol JSON and aggregate CSV for machine-readable inputs. Individual customer IDs are excluded.